

Haychan Shin

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EDUCATION

University of Cambridge, Gonville and Caius College

October 2020 - June 2026

MEng, BA Electrical Engineering and Information sciences

- **Proposed Master's Thesis:** 3D reconstruction of Monocular endoscopic images using Deep Learning: Explored geometric and learning based surface reconstruction of uncalibrated monocular endoscopic images using learning based reconstructive methods such as SfM, MVS, NeRF, 3DGS, VIT (DUST3r)
- **Relevant Modules:** linear algebra, probability theory, object-oriented programming, statistics, stochastic process, MLE, information theory, data transmission, signals and systems, computer vision, probabilistic machine learning

SKILLS

Languages: Python, JavaScript, Octave, C++, HTML, CSS, SQL

Frameworks & Technologies: PyTorch, Tensorflow, HuggingFace, NumPy, pandas, Matplotlib, Seaborn, Flask, PyQt5, FastAPI, Git, Docker, Postman, MongoDB Atlas, Motor, SQLite,

IT: SolidWorks, Altium Designer, Eagle, Linux, KiCAD, LTSpice, Microsoft Excel

WORK EXPERIENCES

3D Reconstruction Researcher | Republic of Korea Army

December 2023 - May 2025

(Python, Pytorch, FastAPI, Neural Radiance Fields, 3D Gaussian Splat)

- Designed a terrain mapping platform using Neural Radiance Fields to get improved depth estimations under varying lighting conditions from multi-view satellite imagery.
- Published novel deep-learning pretraining scheme for enhanced robustness of implicit neural representation under minimal training data, achieving 10% improved SSIM scores.

Electrical Engineering Intern | TTP plc

July 2023 - September 2023

(Python, Matplotlib, PCB Design, Altium Designer)

- Spearheaded the development of a dynamic ray-traced model in Python for a novel optical alignment system culminating in **60% improvement in detection area** and **15% improved precision**.
- Designed, populated and tested PCB designs for sensor arrays, evaluating accuracy and optical reflection properties.

Data Analyst Team Lead | Cascade

July 2021 - September 2021

- Developed backend for a Holiday Carbon Calculator app utilising Google Maps, Hotels, and Skyscanner API to estimate users' carbon footprints for various transportation and accommodation options based on historical and real time data.

PROJECTS

Shazam Clone Music Identifier

(Python, Scipy, Librosa, MongoDB)

- Developed a music recognition system using audio fingerprinting and spectrogram analysis to match short audio clips with songs in a song database, implemented peak energy detection and hash generation for O(1) lookup efficiency.

LLM Behavioural Interview Simulator

(Python, Javascript, HTML, CSS, FastAPI, Postman, Langchain, MongoDB)

- Developed a mock interview simulator leveraging OpenAI Whisper 'Voice-to-Text' and a prompt engineered Gemma 3 to simulate behavioural mock interviews.
- Utilised Retrieval-Augmented-Generation for resumé and web scraped job listings to generate adaptable behavioural questions and follow up questions.

Brain Tumour Semantic Segmentation

(Python, Pytorch, U-net, Medical Imaging)

- Implemented a 3D U-Net on the Brain Tumour Segmentation (BraTS) dataset, leveraging multimodal MRI data to segment and classify Brain tumours for evaluation using Dice Coefficient Loss, Focal Loss for optimal model training

GramifyMyMusic

(Python, Flask, Git, HTML, CSS, Javascript)

- Developed a web app converting user's top artists and songs in a given timeframe into downloadable posters using the Spotify OAuth2 API Module utilising Python, Flask framework and Javascript.